Towards engineering intelligent systems

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This paper sets the rational and historical background for this issue of the Journal. It also provides editorial overviews of the three artificial intelligence (AI) technology themes of 'constraint optimisation and scheduling', 'software agents' and 'soft computing' that are covered by the papers in this issue.

1. Introduction

This themed edition of the Journal is a successor to two earlier special issues. The first, 'Advanced information processing techniques for resource scheduling and planning', published in 1995, and the second, 'Intelligent software systems', published in 1996, presented concepts and demonstrators of intelligent planning and scheduling, and software agents and soft computing respectively.

The work reported in the earlier two special issues has progressed significantly from concepts and demonstrators towards truly intelligent software engineering tools, techniques and solutions to real problems. This mirrors the typical evolution path of any technology that usually proceeds in three phases. The first phase involves a thorough exploration of the technology in order to identify its potential benefits. New methods and techniques are identified and defined resulting in demonstrators for point solutions. The 1995 and 1996 issues reported on work primarily in this first phase of evolution. The second phase of technology evolution involves consolidation, partitioning and methodological rationalisation, wherein understanding of the technology develops to the extent that methodological application of 'tried-and-tested' techniques for well-defined partitioned problems is possible. In this phase lies the genesis and praxis of engineering, and a move from 'point solutions to point problems' to generic problem classes and their solutions. The majority of the papers in this edition fall into this second phase of evolution. The third and final evolution phase comprises routine usage of the technology and standardisation.

The work reported in this issue falls within all the three phases of technology evolution. The papers on soft computing and machine intelligence clearly describe work belonging to the initial concept exploration phase, and demonstrate our vision and pioneering research in building intelligent systems which ensures BT a dominant position in the global information society of the next millennium. The work reported in the other papers can generally be characterised as being of the second phase. However, some of the reported works are already benefiting BT immensely, and can therefore be argued to be already in the third phase. For example, one paper describes the integration of artificial intelligence (AI) constraint optimisation techniques within a dynamic scheduling system central to Work Manager — BT's workforce scheduling system which plans the activities of thousands of technicians all over the UK. This work was awarded a 1997 British Computer Society Information Technology (IT) medal for business and technical contributions. Other papers report on work that is not yet as mature but is already having significant impact. For example, one paper describes BT's ZEUS tool-kit for rapidly engineering certain classes of agent applications. This work won the prestigious 1997 British Computer Society award for IT excellence and innovation, which in turn clearly reinforces BT's brand as an innovative IT company. Some of our work on agent technology is already getting into its third phase with BT playing a vanguard role in the standardisation of agent technology as is described by another paper in this issue.

In the rest of this paper we briefly review the reported work in this BTTJ issue on the three AI technology themes of constraint optimisation, software agent and soft computing respectively.

2. Constraint optimisation

The AI technology of constraint optimisation covers two disciplines — combinatorial optimisation and constraint-based reasoning. Combinatorial optimisation has a long tradition in operations research. In recent years, heuristic search has drawn much interest in the scientific community because of its ability to address in an efficient and natural way the most difficult combinatorial optimisation problems. In contrast with traditional techniques such as Branch and Bound, the inaccuracy of heuristic search (i.e. solutions are not guaranteed to be optimal) is largely compensated for by its efficiency and scalability (i.e. good-quality solutions are obtainable in acceptable time on large-scale problem instances).

Constraint-based reasoning is comparatively a young field; it appeared in the early seventies to become one of the most popular and successful paradigms in AI today. Constraint-based reasoning addresses the problem of satisfying constraints on a set of discrete decision variables. Much work has been dedicated to tackling the intractability of constraint satisfaction problems (finding a complete assignment of the variables that satisfy all the constraints may require a time exponential in the size of the instance) through the design and development of hybrid search algorithms and constraint propagation techniques or the identification of tractable problem classes. Thanks to high-level software engineering, constraint programming tools have now emerged as the most competitive alternatives to address a wide range of industrial problems including resource allocation, scheduling, time-tabling, planning, diagnosis, design and configuration.

Our interest in constraint optimisation is motivated by the simple fact that most real-life decision problems are hybrid and require both constraint-based reasoning and optimisation capabilities from decision-support tools. Our ambition is to integrate both approaches in a single modelling and problem-solving framework. The papers published in this section describe solutions we have developed for addressing different BT problems. The emphasis is put on engineering aspects of problem modelling, algorithmic principles and system integration.

The first paper, by Lesaint et al (p 16), describes an example of integration of heuristic search and constraint programming to address BT's workforce scheduling problem. The output of this work was the dynamic scheduling system that now equips Work Manager and is used all over the UK. This system has received various awards including the 1997 British Computer Society IT medal for its business and technical contribution.

The next paper, by Voudouris, Azarmi and Jones (p 30), presents a solution to the problem of generating and sequencing calls to test BT's billing system. This system, which has also been deployed by BT, provides another proof of the scalability and robustness of constraint optimisation techniques.

The paper by Elfe, Freuder and Lesaint (p 38) introduces a constraint-based reasoning approach to the problem of detecting and resolving feature interaction between telecommunications services at specification time and at run time. This work illustrates the power of constraint modelling but also stresses the need to enhance the knowledge representation capabilities further in order to tackle increasingly complex problem requirements.

The last paper of this technology theme, by Voudouris (p 46), presents and illustrates the versatility of a powerful optimisation technique — Guided Local Search — the success of which on classical problems has been largely demonstrated. This paper provides a good insight into the state of the art in combinatorial optimisation techniques.

Each paper highlights, for a specific application domain, the benefits constraint optimisation technology can provide. Some examples are given below.

- Exact solution modelling — how realistic are the solutions?
- High-quality solutions — how satisfactory are the solutions with respect to predefined targets such as cost minimisation, customer satisfaction, resource utilisation, etc?
- Solution integrity — how well do solutions comply with domain constraints and business rules?
- Efficiency — how much time and resources are needed to generate solutions?

Building on our successes, we are currently undertaking research into the design of generic constraint optimisation components that could be easily and quickly reused to tackle various application domains.

3. Software agents

At the time of the previous themed issue of the Journal on 'Intelligent software systems', software agent technology was in its second infancy — the first followed its birth from distributed artificial intelligence research in the late 1970s and early 1980s. During this period, the field of software agents was less of a technology and more of a research science centred in university laboratories. The primary research concerns were aimed at resolving theoretical issues such as devising reasoning strategies for different contexts.

The second infancy of software agent technology, which spanned the late 1980s into the 1990s, saw a shift in the focus of agent research from scientific concerns to practical and technological concerns, for example, how to build agent systems to address practical problems. With this shift came the notion of agents as embodied intelligences, which, coupled with the idea of users delegating behaviours to agents, proved to be very appealing. In essence, a user's agent was now seen as an intelligent extension of the user — a sociologically and physiologically attractive notion to users. This second infancy of software agent technology also saw the birth of significant over-expectations of the potential impact of the technology on everyday life. To a large extent, this was due to two main factors. Firstly, the exponential and largely unpredicted growth of the Internet implied world-wide interconnectivity and almost instant access to a myriad of Internet resources. Secondly, the concept of mobile agents that roam the Internet, working on behalf of their users simultaneously raised excitement and trepidation in people. However, the reality of the slow rate of technological progress, and the innate difficulty of some of the problems agent researchers are trying to address have served to dampen some of the excessive expectations.

Our previous special issue, published during the second infancy period of agent technology, explored the rapidly evolving area of software agents with some definitive introductory papers, and papers that described concept demonstrators of agent-based systems in a number of application domains. Last year, in a paper reviewing the research and development challenges for agent-based systems[1], we argued that software agent technology had matured to the second phase of technology evolution mentioned earlier. Furthermore, we argued that what was then lacking was the development of methodological approaches for engineering agent-based systems.

The software agents technology section of this special issue describes work we have undertaken to tackle the challenge of developing methodological approaches for engineering agent-based systems. The emphasis of each paper is naturally different, but to a large extent, all the papers focus on addressing problems within the context of facilitating the transition of agent technology from laboratories into widespread industrial use. The first six papers primarily address collaborative agent technology, whereas the final three papers are concerned with personal/ information agent technology.

The paper by O'Brien and Nicol (p 51) discusses the work of FIPA (Foundation for Intelligent Physical Agents), an international standardisation body that aims to define agent interfaces in order to support interoperability between different agent systems. Necessarily, some form of interoperability standard is needed for widespread commercialisation and exploitation of agent technology.

The next paper by Collis et al (p 60) describes the ZEUS collaborative agent building tool-kit. The authors argue that the agent development process should be transformed from a scientific discipline into an engineering one, where developers do not need to know all the intricate science of agent behaviour in order to build working agent systems. They approach the problem through the provision of a library of agent-level components and an environment to support the agent building process. The ZEUS tool-kit won the prestigious British Computer Society 1997 award for technological excellence.

[1]Ndumu D T and Nwana H S: 'Research and development challenges for agent-based systems', IEE Proceedings, Software Engineering, 144, No 1, pp 2—10 (1997).

The next paper, by Ndumu, Collis and Nwana (p 69), describes the technical challenges involved in integrating partial and distributed information sources and services (available from a medium such as the Internet) into a comprehensive integrated value-added service. It then goes on to describe a collaborative agent-based demonstrator travel assistance system that was built using the ZEUS tool-kit. The system combines services from a variety of sources to generate a transatlantic trip itinerary for the user.

The paper by Judge et al (p 79) describes the use of agent technology to enhance a commercial workflow management system. They argue that the agent layer introduced by their approach contributes a much needed flexibility to the workflow management process, wherein the overall system is better able to dynamically react to changing circumstances such as exceptions or new policy decisions. They argue, in effect, that using agents to manage workflow systems provides an 'early win' alternative with near-term benefits, while the much harder and longer-term goal remains the full-scale commercialisation of agent-based process management (the concept of agent-based business process management was presented in our earlier issue).

The paper by Davison, Hardwicke and Cox (p 86) discusses the application of agent technology to the performance management of asynchronous transfer mode (ATM) telecommunications networks. The authors argue that in order to build telecommunications network management systems that can evolve at the same pace as the underlying network infrastructure, the strengths of agent technology in terms of interoperability of heterogeneous systems, loose component coupling and decentralised control are important. As such, they argue the network management function is best performed with simple agents that are dedicated to single tasks and that negotiate with one another, with no pre-built control relationships. They elucidate their ideas with a demonstrator system that manages a network to avoid unnecessary connection rejection when there is spare capacity in the network.

The last paper on collaborative agent technology, by Lee et al (p 94), focuses on the often-neglected questions of performance, scalability and stability of multi-agent systems. These non-functional issues are nonetheless of paramount concern if multi-agent systems are to be commercially deployed on a large scale. How do we measure the performance of a multi-agent system? Further, how do we guarantee certain performance bounds, or worse yet, guarantee that the technology scales up and will not be prone to instability problems? Firstly, however, what do these terms mean in the context of multi-agent systems? The paper is a fledgling attempt at addressing these important questions, and uses example multi-agent systems built using the ZEUS tool-kit to illustrate how these non-functional concerns can be investigated for practical systems.

The paper by Davies, Stewart and Weeks (p 104) is the first of three on personal/information agent systems. It describes a system that encourages and enables the sharing of explicit codified knowledge and tacit knowledge among groups of users. The knowledge shared between the users originates from a variety of sources, including the World Wide Web, an organisation's intranet, or even from other users. The system is another example of a more mature, sophisticated and in-use version of an earlier demonstrator described in our previous special issue in 1996.

The paper by Soltysiak and Crabtree (p 110) discusses the problem of automatically learning a user's interests and preferences in order to provide personalised services to the user. They describe a series of experiments aimed at discovering whether a user's interests could be automatically classified through the use of several heuristics. They conclude that for effective profiling, the process needs to be both data-driven (or bottom-up), and goal-driven (or with top-down guidance from the user).

The final paper of this software agents technology theme by Crabtree and Rhodes (p 118) briefly overviews the emerging field of wearable computing. They argue, first that a wearable computer is not simply a computer that a user wears, it is a host for a set of applications. They describe potential application scenarios for wearable computers, along with their design needs. Finally, they describe their experiences with a wearable reminder agent.

It is hoped that the selection of agent-related papers in this special issue demonstrates the advancing maturity of our R&D in this field. There is still some way to go yet before agent-enhanced systems become commonplace in society; however, the evidence points unerringly to the inevitability of that day.

4. Soft computing

The philosophical argument for soft computing (SC) is that humans deal with uncertain and imprecise information everyday and are remarkably consistent in processing such information. Thus, the primary aim of SC research is to develop hybrid intelligent systems that exploit the tolerance of imprecision, uncertainty and partial truth associated with almost every aspect of real world problems. Such systems will possess the much-needed flexibility that is the key to providing reliable human-centred systems that work for, with or on behalf of humans. The soft computing section of this issue describes exploratory work we have undertaken to investigate the use of soft computing techniques for developing intelligent systems that support such flexible human/machine interaction.

The paper by Azvine and Wobcke (p 125) is an overview of the challenges and opportunities for research and development in human-centred intelligent systems. It offers a conceptual architecture for building such systems and highlights the important components of such an architecture from the point of view of flexible processing. The paper then describes the contribution of soft computing techniques to enhance the flexibility of the proposed human-centred intelligent systems.

The next three papers present three application areas where a system's flexibility plays a key role in determining how its users' perceive the system. The paper by Tsui et al (p 134) describes the architecture of a distributed software platform of an intelligent multi-modal interface used to obtain information from a large workforce management system. The architecture supports the integration of speech, text, simulated gestures and gaze input, allowing for a more natural and flexible user/system interaction.

The paper by De Roeck et al (p 145) on an intelligent directory enquiry assistant describes a language-engineering project that combines the latest natural language processing techniques and information retrieval. The paper describes the development of a prototype system for accessing classified directory information that goes beyond the capabilities of present day systems which rely on simple pattern matching or database look-up. The paper also reports on testing the prototype system using an existing classified directory database.

The next paper, by Baldwin, Case and Martin (p 156), on machine interpretation of facial expressions, describes an attempt at producing a system that can infer human mood from visual clues. It describes a system for detecting facial features concentrating on the shape of the mouth and the eyes. They have proposed a fuzzy approach to improve the classification of facial features using colour, intensity and positional information. The authors have verified experimentally that their methods are reliable, accurate and robust.

The next paper, by Baldwin, Martin and Azvine (p 165), is a survey concentrating on the need for, and the application of soft computing techniques in intelligent knowledge-based systems. It makes a strong case for the necessity to handle uncertainty in knowledge-based systems at the input/output and the processing stages. Logic programming is presented as a powerful framework for representing and manipulating knowledge in intelligent systems when uncertainty in relations, inference and fusion of uncertain conclusions plays a central role. The paper then concentrates on a number of tools that can be used to build intelligent knowledge-based systems including Fril, which uses the mass assignment theory. The authors conclude by making a case for assessing the various available tools using a number of benchmarking problems.

The last of the themed papers, by Nauck and Kruse (p 180), describes a soft computing tool that can be used to learn fuzzy classification rules from examples. It describes the tool in terms of its model, platform, capabilities, learning algorithm and the interface. The paper concludes by pointing to practical applications areas of the tool in medical diagnosis, stock market analysis and risk assessment.

5. Conclusions

Intelligent systems refer to a class of software systems that use artificial intelligence (AI) techniques in order to carry out tasks which would be considered 'intelligent' if done by humans. Such systems will extend, multiply and leverage our mental and physical abilities and pervade all aspects of our lives. The papers in this issue demonstrate the vision, breadth and depth of the intelligent systems research and development programme within BT's Applied Research and Technology department, concentrating on AI technologies of constraint optimisation and scheduling, software agents, and soft computing.

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